



Improving fertilizer management in the U.S. and Canada for N₂O mitigation: Understanding potential positive and negative side-effects on corn yields



Diego Abalos^{a,*}, Simon Jeffery^b, Craig F. Drury^c, Claudia Wagner-Riddle^a

^a School of Environmental Sciences, University of Guelph, Guelph, ON N1G 2W1, Canada

^b Department of Crop and Environment Sciences, Harper Adams University, Newport, Shropshire TF10 8NB, UK

^c Harrow Research and Development Center, Agriculture and Agri-Food Canada, Harrow, ON N0R 1G0, Canada

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ABSTRACT

The potential for mitigation of N₂O emissions arising from fertilizer management practices in corn systems has been scientifically assessed in recent decades. Given the strong association between fertilizer management and crop productivity, which to a large extent determines farmers' willingness to adopt such practices, it is essential to incorporate the impacts on yields before any mitigation practice can be recommended. Here, we address this issue by integrating available results via meta-analysis to quantitatively evaluate the effect of fertilizer management practices on N₂O emissions and corn yields in the U.S. and Canada. The combined application of nitrification and urease inhibitors represented the best N₂O mitigation option of the practices tested, and their use tended to lead to concurrent increases in crop yields. In comparison, polymer-coated urea was less effective at both reducing N₂O emissions and increasing corn yields. Substitution of synthetic fertilizers by organic sources served to reduce N₂O emissions but at the cost of significant yield penalties. Sidedress fertilizer applications rather than applying all the fertilizer at planting consistently increased crop productivity. No significant effects were found for the use of urea vs. ammonium nitrate or urea ammonium nitrate, nor for fall vs. spring fertilizer application. However, further research is needed to confirm these observations due to the highly variable nature of the data. Two key soil parameters were identified: 1) Soil texture; N₂O and crop yield are highly responsive to fertilizer management in fine-texture soils; and 2) C/N ratio; mitigation of N₂O emissions without concurrent yield reductions is most frequently achieved in soils with a low C/N ratio (*i.e.* <12.5). Our study provides a step forward in achieving an integrated understanding of the positive and negative side-effects of fertilizer management practices for N₂O mitigation in agriculture.

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1. Introduction

Nitrous oxide (N₂O) is a potent greenhouse gas with a global warming potential 298 times higher than carbon dioxide (CO₂) on a per mass basis, and is the largest stratospheric ozone-depleting substance (IPCC, 2007; Ravishankara et al., 2009). Agricultural soils are by far the main anthropogenic source of N₂O emissions, primarily as a result of the addition of synthetic nitrogen (N) fertilizers and animal manures to soil (Bouwman et al., 2002). An intensive research effort over recent decades has enabled the identification of a number of fertilizer management options capable of reducing N₂O emissions (Millar et al., 2010).

Nonetheless, there are large uncertainties associated with the mitigation potential of these practices due to the complex interaction of the biotic and abiotic factors that regulate soil N₂O emissions (Venterea et al., 2012).

The level of implementation of all mitigation practices in agriculture depends to a large extent on their effect on crop productivity, as this will determine farmers' willingness to adopt these practices. Food security may be hampered if this issue is not comprehensively addressed. Yet, the potential consequences of N₂O mitigation practices on crop yield remain largely unexplored (Millar et al., 2010). Both positive and negative effects, depending on the practice, can be expected. For example, simultaneous N₂O reductions and crop yield enhancements following the application of new fertilizer technologies have been reported in the scientific literature (e.g. Liu et al., 2013). Conversely, other studies reported lower yields with fertilizer types that were successful at mitigating N₂O

* Corresponding author.

E-mail addresses: diego.abalos@upm.es, diegoabalosr@gmail.com (D. Abalos).

emissions (e.g. Halvorson et al., 2010a; Gagnon and Ziadi, 2010; Gagnon et al., 2011; Fujinuma et al., 2011). In order to effectively guide policy regarding the dual goals of food security and climate change mitigation it is crucial to gain a quantitative understanding of the potential trade-offs between N₂O emissions and crop yield.

The Corn Belt in the Midwest U.S. and central Canada is one of the world's most important food and fiber producing areas, representing a unique biophysical area with large inherent capacity to generate high N₂O emissions. This is because corn receives more N than any other U.S. crop (based upon the combination of high application rates with large land areas) and the soils are rich in organic matter and with predominantly medium to heavy texture sustaining high water-holding capacity (Jarecki et al., 2015). In a recent paper, Decock (2014) quantitatively investigated the effects of a suite of agronomic management practices on N₂O emissions in corn cropping systems from this area. However, the effects of those practices on crop yield were not reported in the study. This lack of information constrains the wide-scale application of N₂O mitigation practices as most corn farmers in the U.S. will not adopt these strategies while their effect on yield is uncertain (Stuart et al., 2014).

Here, we integrate available results via meta-analysis to quantitatively evaluate the effect of fertilizer management practices on N₂O emissions and corn yields in the U.S. and Canada. This approach allowed us to assess trade-offs where yield penalties may represent a potential barrier to implementation of an available mitigation option and to highlight practices where N₂O mitigation practices can function alongside positive yield effects. Further, to improve our understanding of the main drivers underlying observed effects, we investigate response variables as a function of key soil factors which are likely to govern both N₂O emissions and crop yields.

2. Materials and methods

2.1. Data search and selection criteria

A meta-analysis was conducted to concurrently characterize the response of N₂O emissions and crop productivity to a set of fertilizer management options. The evaluated options were: *Synthetic vs. Organic* fertilizers; *Fertilizers without urease and nitrification inhibitors vs. Fertilizers with inhibitors*; *Urea vs. Polymer-coated urea (PCU)*; *Urea vs. Ammonium Nitrate (AN) or Urea Ammonium Nitrate (UAN)*; *Fall vs. Spring* fertilizer application; and *Planting vs. Sidedress* fertilizer applications. Only studies reporting side-by-side comparisons of specific practices were used. For example, data were extracted from studies where a fertilizer without inhibitors application (control) could be compared to an equivalent treatment with inhibitors with all other factors unchanged. In order to determine the key soil drivers affecting the response variables, the experiments were grouped in terms of soil texture, soil pH, soil organic carbon (SOC) and soil C/N ratio. Data were grouped to maximize in-group homogenization and to facilitate cross-comparisons between studies (Abalos et al., 2014). Soil texture was grouped into three basic classes (fine, medium and coarse) according to Soil Survey Staff (1999). Soil pH was divided into three common ranges (≤ 6 , 6–7 and ≥ 7). For SOC, data were grouped according to three groups (< 20 ; 20–40; > 40 g C kg⁻¹). Soil C/N ratios were divided into two groups according to critical soil C/N ratios within the context of N mineralization and immobilization, as described by Hodge et al. (2000) (low: < 12.5 ; intermediate to high: > 12.5).

Two datasets were compiled, one for N₂O emissions and one for crop yield. A literature search using Google Scholar and Web of Science was conducted with the keywords (“nitrous oxide”, “N₂O”, “corn”, “maize”, “yield”, “productivity”, “Corn Belt”). The cut off for

Table 1
Description of the studies used in the meta-analysis.

Reference	Comparison	Peer-reviewed	Available data	
			N ₂ O	Yield
Adviento-Borbe et al., 2010	<i>Synthetic vs. Organic</i>	Yes	X	X
Chantigny et al., 2010	<i>Synthetic vs. Organic</i>	Yes	X	
Chantigny et al., 2013	<i>Synthetic vs. Organic</i>	Yes	X	
Drury et al., 2012	<i>Planting vs. Sidedress; U vs. PCU</i>	Yes	X	X
Duxbury et al., 1982	<i>Synthetic vs. Organic</i>	Yes	X	
Fan et al., 1997	<i>Planting vs. Sidedress</i>	Yes	X	
Gagnon et al., 2012a	<i>Synthetic vs. Organic</i>	Yes		X
Gagnon et al., 2012b	<i>With vs. Without inhibitors</i>	Yes		X
Halvorson et al., 2010a	<i>With vs. Without inhibitors; U vs. PCU</i>	Yes	X	X
Halvorson et al., 2010b	<i>With vs. Without inhibitors; U vs. PCU</i>	Yes	X	X
Halvorson et al., 2011	<i>With vs. Without inhibitors; U vs. PCU; U vs. AN/UAN</i>	Yes	X	X
Hatfield and Parkin 2014	<i>With vs. Without inhibitors</i>	Yes		X
Hernandez-Ramirez et al., 2009	<i>Synthetic vs. Organic; Fall vs. Spring</i>	Yes	X	
Jagadamma et al., 2009	<i>Synthetic vs. Organic</i>	Yes		X
Jarecki et al., 2009	<i>Synthetic vs. Organic</i>	Yes	X	
Kariyapperuma et al., 2012	<i>Synthetic vs. Organic</i>	Yes	X	
Ma et al., 2005	<i>Planting vs. Sidedress</i>	Yes		X
Nelson et al., 2009	<i>U vs. PCU</i>	Yes		X
Paniagua, 2006	<i>U vs. PCU</i>	No	X	X
Parkin and Hatfield 2010	<i>With vs. Without inhibitors</i>	Yes		X
Paul and Beauchamp 1993	<i>Synthetic vs. Organic</i>	Yes		X
Rochette et al., 2004	<i>Synthetic vs. Organic; Fall vs. Spring</i>	Yes	X	
Rochette et al., 2008	<i>Synthetic vs. Organic</i>	Yes	X	X
Roy et al., 2014	<i>Planting vs. Sidedress</i>	Yes	X	X
Schwager, 2015	<i>Fall vs. Spring</i>	No	X	X
Sistani et al., 2010	<i>Synthetic vs. Organic</i>	Yes	X	
Sistani et al., 2011	<i>Synthetic vs. Organic; With vs. Without inhibitors; U vs. PCU; U vs. AN/UAN</i>	Yes	X	X
Smith et al., 2002	<i>Synthetic vs. Organic</i>	Yes	X	
Venterea et al., 2011	<i>With vs. Without inhibitors; U vs. PCU</i>	Yes	X	X
Wagner-Riddle et al., 2007	<i>Planting vs. Sidedress; U vs. AN/UAN</i>	Yes	X	X
Zebbarth et al., 2008	<i>Planting vs. Sidedress</i>	Yes	X	

published articles to be included in the databases was the end of March 2015. Literature cited in the articles was also reviewed to complement the keywords search. To be considered for inclusion; the studies had to meet the following criteria: (1) contain sufficient information about the experimental design; (2) be a field experiment; (3) include a minimum of three replicates; and (4) report a measure of variance. For the N_2O dataset an additional criteria was added: (5) have utilised a suitable sampling methodology to robustly account for temporal and spatial variations. In addition to published papers; two studies from the gray literature were included (Paniagua, 2006; Schwager, 2015) to reduce the potential artifacts caused by publication bias (Rosenthal and Rosnow, 1991). The initial databases consisted of 572 pairwise comparisons from 55 articles for N_2O emissions; and 726 pairwise comparisons from 50 articles for yield. However, several papers did not meet the inclusion criteria due to reporting insufficient information on auxiliary variables or due to containing confounding factors or controls that were inappropriate for this analysis. Therefore, in total, 23 studies and 200 observations were used for N_2O ; and 20 studies providing 153 observations were used for yield (Table 1).

When data were only provided in graphic format, Plot Digitizer 2.6.6 (Huwaldt, 2014) was used to extract data. Standard deviation (SD) was used as a measure of variance, which was calculated from the published measure of variance in each study if necessary. Efforts were made to contact the corresponding authors if no measure of variance was provided in the study. In most cases these data were provided by the researchers. When a measure of variance was not available, it was estimated as being 150% of the mean of all the available SDs (Lubbers et al., 2013). The inverse of the pooled variance was then calculated using the obtained SD value.

2.2. Fertilizer management strategies

For the *Synthetic vs. Organic* fertilizers category, the synthetic sources were calcium ammonium nitrate, ammonium nitrate, urea ammonium nitrate and urea; the main organic fertilizers included were solid and liquid manures (treated and untreated), biosolids and compost from different sources. For the *Fertilizers without urease and nitrification inhibitors vs. Fertilizers with inhibitors*, the inhibitors used were a combination of NBPT (*N*-(*n*-butyl)thio-phosphoric triamide; urease inhibitor) and DCD (dicyandiamide;

nitrification inhibitor) for all cases except for [Parkin and Hatfield \(2010\)](#), where only the nitrification inhibitor nitrapyrin was used. The enhanced efficiency fertilizer employed in all studies for the *Urea vs. Polymer-coated urea* group was “Environmentally smart nitrogen” (ESN). The application timing for the *Planting vs. Sidedress* category was within 5 days after planting and at the V6 stage for sidedress, except for [Roy et al. \(2014\)](#) where the fertilizer was applied at the V8 stage, and [Hatfield and Parkin \(2014\)](#) where 4.5 kg N ha^{-1} were applied at the V12 stage in addition to the V6 fertilization.

2.3. Data analysis

Effect size was used as the metric for analysis, which was obtained through the natural log-transformed response ratio ([Hedges et al., 1999](#)):

$$\ln R = \ln \left(\frac{\bar{x}^T}{\bar{x}^C} \right)$$

where $\bar{x}^T$ is the mean of cumulative N_2O emissions or crop yield in the treatment group, and $\bar{x}^C$ is the mean of cumulative N_2O emissions or crop yield in the control group.

All analyses were carried out in MetaWin 2.1 ([Rosenberg et al., 2000](#)). A categorical random effects model was used where the estimated pooled variance was ≥ 0 ; a fixed effects model was used for the remaining cases. Confidence intervals (95% CIs) were generated by bootstrapping (9999 iterations). If the 95% CIs did not overlap zero, mean effects were considered significantly different from zero. If categorical grouping 95% CIs did not overlap, they were considered significantly different from each other ([Hedges et al., 1999](#)). To facilitate interpretation, the response ratios were exponentially transformed and reported as percentage change of the treatment relative to the control ($[R - 1] \times 100$).

Rosenthal's Fail-safe N technique at $P = 0.05$ was used to test the potential impact of publication bias ([Rosenthal and Rosnow, 1991](#)). The Fail-Safe N for all the fertilizer management practices showing significant effects on N_2O emissions or crop yields was always lower than the critical value suggested by Rosenthal ($5n + 10$, with n as the number of original studies included in the meta-analysis). Thus, it is unlikely that publication bias existed in the literature to a sufficient extent to affect the overall statistical significance of the results and hence, nor the interpretation in our analysis.

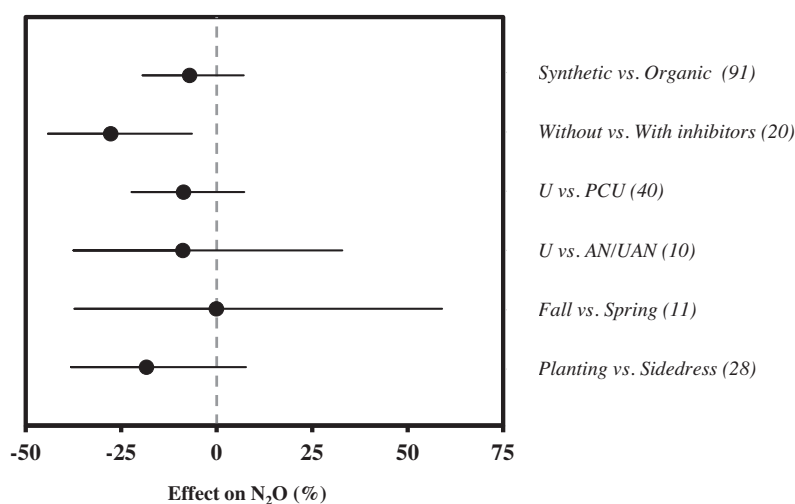


Fig. 1. Effect of fertilizer management practices on N_2O emissions (%). Mean effect and 95% confidence intervals are shown. Sample sizes (i.e. the number of control–treatment pairs) are shown in parentheses.

3. Results

3.1. Overview of the databases

The studies included in the databases were predominantly from the U.S. Corn Belt, stretching from the states of Colorado to New York, and from the provinces of Ontario and Quebec, which account for the vast majority (94%) of corn production in Canada (Yang et al., 2013). There were studies from the three Corn Belt ecoregions defined by the U.S. Environmental Protection Agency (i.e. the Eastern, Central and Western Corn Belt Plains). All the articles were published between 1982 and 2014. Most articles on N₂O used chambers for gas sampling and were conducted over less than one year; two to three years was the most frequent duration of experiments reporting yield. The mean fertilizer-N application rate was 165 kg N ha⁻¹ with a SD of ± 51 kg N ha⁻¹, which is within the normal range for the studied regions (Bierman et al., 2012).

3.2. Nitrous oxide emissions

Fig. 1 shows the mean effect size for N₂O emissions grouped by fertilizer management practice. Only the combined application of urease and nitrification inhibitors led to a significant N₂O decrease (mean effect: -27.7% ; CI: -44.2 to -6.5%). The average effect of all the fertilizer management practices evaluated was a significant N₂O reduction of -11.4% (data not shown). No other categories showed a statistically significant effect although there was a general trend for treatment means to show a reduction compared to controls. Data on N₂O fluxes are notoriously variable, and this high variability was further confounded by the low number of pairwise comparison included in the *Fall vs. Spring* and *U vs. AN/UAN* groups. This precludes robust interpretation of the likely impact of fertilizer application on N₂O fluxes for these categories.

3.3. Crop yield

In terms of N fertilization strategy effects on crop yields, applying fertilizer as a sidedress led to a statistically significant yield increase of 4% compared to fertilizer applied at planting (Fig. 2). The use of organic fertilizers led to a significant reduction (11.3%) in crop productivity compared to synthetic fertilizers. None of the remaining fertilizer practices showed statistically significant effects, with all means close to zero effect size, ranging from -0.71% (*U vs. AN/UAN*) to 0.81% (*Without vs. With inhibitors*).

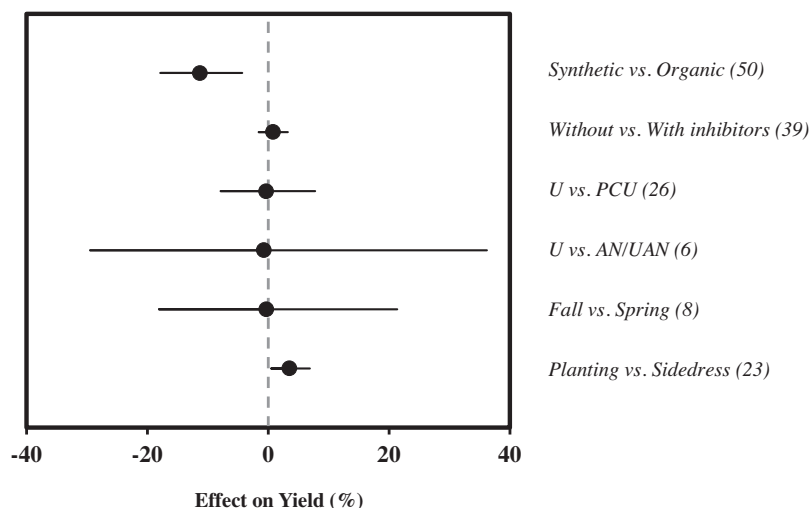


Fig. 2. Effect of fertilizer management practices on corn yield (%). Mean effect and 95% confidence intervals are shown. Sample sizes (i.e. the number of control–treatment pairs) are shown in parentheses.

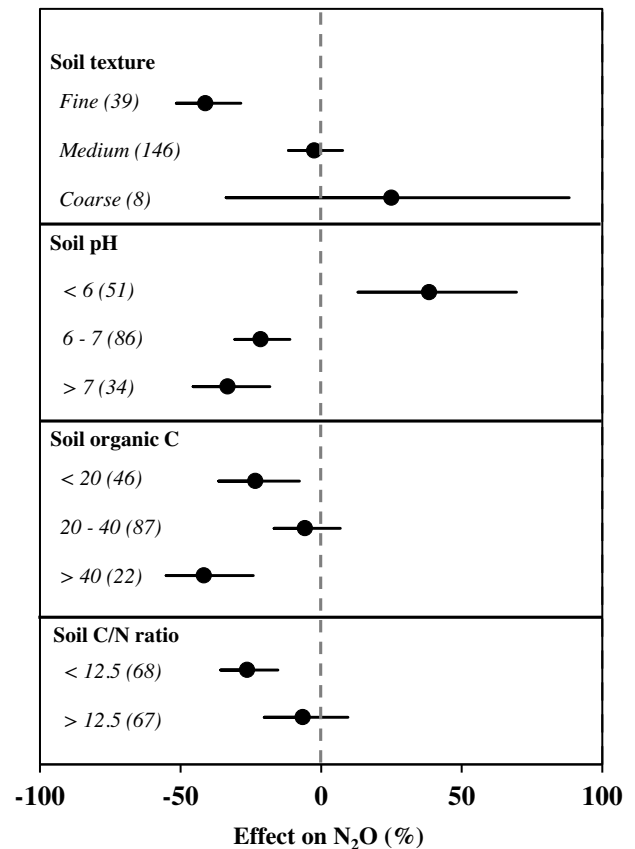


Fig. 3. Effect of fertilizer management practices on N₂O (%) for different soil textures (*Fine*, *Medium*, *Coarse*), soil pH groups (<6, 6–7, >7), soil organic C contents (<20, 20–40, >40) and soil C/N ratios (<12.5, >12.5). Mean effect and 95% confidence intervals are shown. Sample sizes (i.e. the number of control–treatment pairs) are shown in parentheses.

3.4. Soil factors and N₂O emissions

The response of N₂O emissions to the fertilizer options tested was largely determined by soil texture, with the largest N₂O reductions ($-41\% \pm 10\%$) found in fine-textured soils (Fig. 3). Implementation of mitigation practices also led to N₂O reductions in soils of neutral (6–7) to high (>7) pH. Conversely, N₂O emissions

increased significantly in response to the mitigation practices in acidic (<6) soils. Soil organic C was found to interact non-linearly, with statistically significant reductions in N₂O emissions being observed in the <20 and >40 sub-categories; no such effect was observed for the 20–40 sub-category. Significant N₂O reductions were observed in soils with a C/N ratio <12.5, whereas no significant effect of the mitigation practices was observed in soils with a higher ratio.

3.5. Soil factors and crop yield

Adoption of N₂O mitigation practices in soils of medium or coarse texture did not significantly affect corn yields (Fig. 4). However, significant yield reductions were observed in fine-textured soils. Nitrous oxide mitigation practices also led to statistically significant yield reductions in soils with a neutral pH (6–7), soils with a high SOC (>40) and a C/N ratio >12.5. Such practices did not significantly interact regarding yield with any other sub-category of soil factors (Fig. 4).

4. Discussion

4.1. Potential of fertilizer management strategies for N₂O mitigation and side-effects on corn yields

The combined application of nitrification and urease inhibitors was the only mitigation option resulting in a significant N₂O reduction (Fig. 1). Nitrification inhibitors are recommended by IPCC as a potential mitigation option for agricultural N₂O emissions

(IPCC, 2014) and previous meta-analyses suggest that this potential will likely be realised under field conditions (Akiyama et al., 2010; Decock, 2014; Qiao et al., 2015). Here, we provide further evidence of the efficacy of such inhibitors and show that their application does not lead to yield penalties (Fig. 2); rather they tend to generate small yield increases for corn production systems in the U.S. and Canada. Following analysis of studies from across the globe utilising a variety of crops (including two studies and seven pairwise comparisons of corn in the U.S.), Abalos et al. (2014) previously reported that the combined application of these fertilizer technologies tended to enhance crop productivity. Our analyses here comprise eight studies and 39 pairwise comparisons, thereby significantly increasing the confidence in our results for the region and crop under study. By better synchronizing fertilizer N release and crop N demand, these products offer the best option of all those tested in our study to balance N₂O emissions and crop yields. However, because their use represents an additional cost for farmers, the economic viability of these products will likely depend on initiatives such as the implementation of Emission Trading Schemes including greenhouse gas emissions from agriculture, or subsidies from public institutions to cover the cost difference. Our meta-analysis does not indicate whether these technologies allow reduced N rate without affecting yield, which may compensate for their higher costs.

There was no significant difference in terms of N₂O emissions regardless as to whether synthetic or organic fertilizers were applied (Fig. 1), which contrasts with the results reported by Decock (2014). However, in that study the manure N-rate was higher than synthetic N-rate for 22 out of 73 pairwise comparisons. This represents a confounding variable that may explain the differences found between studies. Further, organic fertilizers may reduce N₂O emissions by reducing the N₂O/N₂ ratio (Dittert et al., 2005) and/or by inducing N immobilization (Chantigny et al., 2013) as a consequence of the organic C added. However, using organic N sources instead of mineral fertilizers led to significant yield reductions of c. 11% (Fig. 2). Although organic amendments may deliver a range of ecological and social benefits, their effect on yield must be assessed due to their pivotal importance for sustainable food security (Godfray et al., 2010; Foley et al., 2011). Therefore, our results confirm that the use of organic fertilizers should only be encouraged under those agroecological conditions where they perform best (Seufert et al., 2012) or where other socio-economic and environmental considerations outweigh yield effects as a priority.

Of the management practices tested, sidedress fertilizer application instead of applying all the fertilizer at planting was the only option that significantly increased crop yields (c. 4%; Fig. 2) and, although not statistically significant, showed the second largest mean effect for N₂O abatement (c. 18%; Fig. 1). In-season N applications (from V6 to V8 stages) allow for adjustments due to weather variations, ultimately resulting in higher soil N availability when the crop requires the nutrient most. Higher crop N uptake increases yield and reduces the availability of mineral N sources for the N₂O-producing processes (Abalos et al., 2014) as well as reducing leaching and associated negative environmental effects. Unlike the use of inhibitors, sidedress fertilizer applications do not incur additional costs for farmers (provided the required equipment is available) and thus, implementation of this practice offers a robust opportunity to increase the sustainability of corn production systems in the U.S. and Canada.

Polymer-coated urea (PCU) is an enhanced-efficiency fertilizer (as are urease and nitrification inhibitors) that was developed to increase the efficiency of fertilizer use by crops. Our meta-analysis shows that, compared to the combined application of inhibitors, PCU has a lower potential for N₂O mitigation (Fig. 1) and for crop yield improvements (Fig. 2) as shown by the lack of statistically

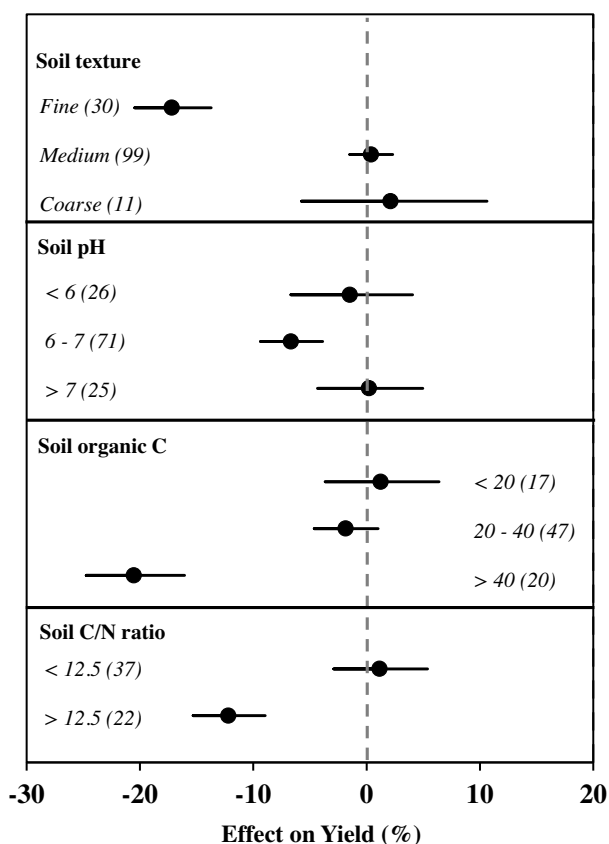


Fig. 4. Effect of fertilizer management practices on corn yield (%) for different soil textures (Fine, Medium, Coarse), soil pH groups (<6, 6–7, >7), soil organic C contents (<20, 20–40, >40) and soil C/N ratios (<12.5, >12.5). Mean effect and 95% confidence intervals are shown. Sample sizes (i.e. the number of control–treatment pairs) are shown in parentheses.

significant response to PCU compared to urea in these agro-ecosystems. Most studies in our meta-analysis were conducted on rainfed corn, where crop water requirements frequently exceed soil water availability (Grassini et al., 2009). Therefore, in rainfed crops soil moisture is usually lower compared to irrigated systems, providing two possible reasons for the observed results: (1) low soil moisture favored nitrification as a source of N_2O and therefore increased the mitigation efficiency of nitrification inhibitors (Akiyama et al., 2010), and (2) limited water delayed the release of N from PCU for a period of time that did not match crop N needs (Shaviv, 2000). Future studies should explore the effects of these products under different soil moisture levels to better understand whether the trends found in our study remain true across a wider range of soil and climatic conditions.

No significant effects were found by replacing urea with ammonium nitrate (AN) or urea ammonium nitrate (UAN), in terms of either N_2O emissions (Fig. 1) or crop yields (Fig. 2). The same is true for the comparison between fall and spring fertilizer application. However, these results are based on large variances confounded by a paucity of reported data. As such, in this analysis it is not possible to determine whether the lack of effect is due to the low number of studies included in those categories (three for *U* vs. *AN/UAN*; three for *Fall* vs. *Spring*; Table 1) or due to the inherent high variance of data reported within the practices analysed in each group. In the U.S. Corn Belt, a substantial amount of N fertilizer is applied during the fall, prior to the growing season, and urea is the most commonly used form of N (Bierman et al., 2012). More studies focused on these practices are necessary to gain a quantitative understanding of their effect on N_2O emissions and crop productivity in order to effectively form policies based on scientific evidence.

4.2. Soil abiotic factors as drivers of the effects on N_2O emissions and crop yield

The average effect for all practices was a significant N_2O reduction, which demonstrates that recent research has identified an array of fertilizer management options with the potential to abate N_2O emissions. However, at the level of single comparisons, five out of six options included in this analysis did not significantly suppress N_2O emissions (Fig. 1). Further, the effect of these practices on corn yield ranged from positive to negative, again with four out of six options included here showing non-significant effects (Fig. 2). To a large extent this is because these management options are associated with large uncertainties caused by the current lack of understanding of the complex interactions between soil factors, as constrained by climate, governing N_2O emissions and crop productivity. These interactions can be synergistic or antagonistic, and may act in opposing ways on N_2O -producing processes and crop growth. Figs. 3 and 4 provide further insights into the role that key soil abiotic factors may play in this issue.

The efficacy of fertilizer management practices on reducing N_2O emissions was highest in fine-textured soils (Fig. 3), which may be explained by soil water dynamics. It is commonly accepted that fine-textured soils favor denitrification and promote higher N_2O emissions due to higher water-holding capacity compared to medium- and coarse-textured soils (e.g. Weier et al., 1993; Bollmann and Conrad, 1998; Pelster et al., 2012). Therefore, this may be a consequence of the largest potential N_2O emissions occurring from these types of soils and hence them having the largest abatement potential. For much of the rainfed U.S. and Canada Corn Belt, soil water and N availability are the most important growth-limiting factors. However, in fine-textured soils, water limitation is comparatively less severe, and thus N availability may become the key factor. This explains why the largest yield penalties were found for these soils (Fig. 4); fertilizer

management options which create asynchronies between N supply and N demand, such as applying organic sources, will have a higher impact on corn grown in fine-textured soils due to this N limitation. Then, these results show that fertilizer options have to be managed with particular care in fine-textured soils, as they are more responsive in terms of both N_2O emissions and yield.

The ultimate goal of intensive and sustainable fertilizer management is to apply N in a way that maximizes crop uptake while minimizing N capture by nitrifiers and denitrifiers and therefore N_2O emissions, among other N losses. According to our results (Fig. 3 and 4), this goal can be best achieved in soils of low C/N ratio (<12.5). Therefore, while it is difficult to assess direct competition between plants and microorganisms for soil N, our analyses suggest that, in soils with a low C/N ratio, crop N uptake is favored over microbial N consumption. This implies that avoiding trade-offs in terms of yield losses may be more challenging in soils with higher C/N ratios, which would require more targeted fertilizer management decisions. The underlying mechanisms for such responses are unclear and warrant further investigation.

Mitigation practices were effective at reducing N_2O emissions in neutral and alkaline soils but not in acidic soils (Fig. 3). Liming may be needed for the latter soils to minimise flux increase and to maximize the efficacy of the abatement potential of these practices while concurrently increasing crop productivity (Pagani and Mallarino, 2015).

Large yield reductions were found in soils with high SOC (Fig. 4), but this effect appears likely to have arisen due to a strong confounding factor in our analyses. Most of the observations included in the high SOC group correspond to the replacement of synthetic fertilizers by organic sources; this also presents the likely explanation for the relatively high SOC levels in this group. Organic amendments may result in yield increases over time when applied to soils of low fertility. However, in soils with inherently high organic matter content, the release of plant-available mineral N from organic sources is often too slow to keep up with the high crop N demand during the peak growing period (Seufert et al., 2012). Thus, evidence on the interactions between SOC and fertilizer practices and the consequences for yield and for N_2O mitigation remain inconclusive.

4.3. Knowledge gaps and limitations of this study

Several practices widely used by farmers in the Corn Belt have received little attention in the scientific literature. For example, only three papers have addressed the differences between shallow and deep fertilizer injection: (1) Drury et al. (2006; 2 vs. 10 cm); (2) Fujinuma et al. (2011; 10 vs. 20 cm); and (3) Maharjan and Venterea (2014; 10 vs. 20 cm). In study (1), higher emissions were measured for the deeper depth injection; the opposite was found in (2) and no effect was observed in (3). Further research is needed to unravel these conflicting results including yield effects. Similarly, a lack of data precluded robust comparisons between broadcast fertilizer application and fertilizer incorporation. The same is true for single vs. multiple side-dress applications. In the study by Adviento-Borbe et al. (2007) multiple side-dress applications induced higher N_2O emissions but also higher yields (for three out of the four possible comparisons). Finally, although the effects of specific practices have generally been studied in isolation, it is the combination of practices that often shows greatest promise for N_2O mitigation without compromising crop yields or resulting in other undesired side-effects (Plaza-Bonilla et al., 2014; Pittelkow et al., 2015). Nonetheless, few studies have compared the simultaneous impact of multiple management options. With such investigations it would be possible to identify interactive effects representative of actual farmers' practices, including additional confounding variables. For example, the timing of N application is

usually confounded with N source and N application method. This is because N application at planting typically uses granular urea that is broadcast and incorporated whereas sidedress N application typically uses liquid UAN which is banded or injected into the soil.

Routine reporting of crop yield data in N₂O emission studies is vital to achieve accurate accounting of N dynamics and budgets through agroecosystems. As highlighted by Van Groenigen et al. (2010), yield data are relatively inexpensive and easy to collect compared with flux measurements. However, data should be reported separately for each variable (i.e. the use of yield-scaled N₂O emissions as unit should be used in addition to, not as replacement of, yield and N₂O data) in order to comprehensively evaluate the specific response of each variable to the treatments under study. Such effort is crucial to increase co-benefits and reduce trade-offs and adverse side-effects of agricultural management practices.

Sustainable agriculture must comply with environmental, economic and social goals, and therefore requires a systems approach. Nitrous oxide emissions and crop productivity only represent two elements of this multidimensional challenge. Trans-disciplinary research is needed so as to further increase our systems knowledge of global issues such as food production and climate change. Further, in order to translate this knowledge into fertilizer recommendations, it is necessary to build a mechanistic framework in which recommendations regarding the application of each practice are provided for the specific environmental and soil conditions that maximize its benefits.

5. Conclusions

The combined application of nitrification and urease inhibitors should be encouraged as it represents the best mitigation option for N₂O emissions of the practices tested, and their use tends to increase crop yield. In comparison, polymer-coated urea seems to be less effective but further research is needed to evaluate this product under a wide range of environmental and soil conditions, especially its dependency on soil moisture conditions. Total or partial substitution of synthetic fertilizers by organic sources may serve to reduce N₂O emissions but potentially at the cost of significant yield penalties. Sidedress fertilizer applications consistently increased crop productivity compared to applying all the fertilizer at planting without raising N₂O emissions. Two key soil factors for the variables under study were identified, providing valuable information from a mechanistic perspective: (1) N₂O and crop yield are highly responsive to fertilizer management in soils of fine-texture; and (2) minimizing N₂O emissions without yield reductions is most frequently achieved in soils of low C/N ratio (<12.5).

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